Unit 1: Functions

Relation: A set of ordered pairs (x, y)

x can be the input and y is the output from the function

What is a function?
- is a relation where each input (x-value) has exactly one output (y-value)

Function Notation
- f(x)
  - f(x) is a replacement for y
  - The function (f) evaluated at x
  - Just a way to describe and name functions for easier communication

Q1) Explain whether they are functions or not
a) { (1, 2), (2, 3), (3, 4) }
b) { (1, 2), (2, 2), (1, 3) }
c) { (x, y) | $y = x^2$ }
d) { (x, y) | $x = y^2$ }
e) { (x, y) | $x^2 + y^2 = 9$ }

f(x) = 3x+1
- f(0)
- f(2)
- f(-3)
- f(a)
- f(x+1)

g(x) = x^2 - 5x + 6
g(1)
g(2)
g(-1)
g(a+1)

h(x) = sqrt(x-2) for which values of x is h(x) defined? (i.e. what is the domain of this fxn)

$\sqrt{x-2}$

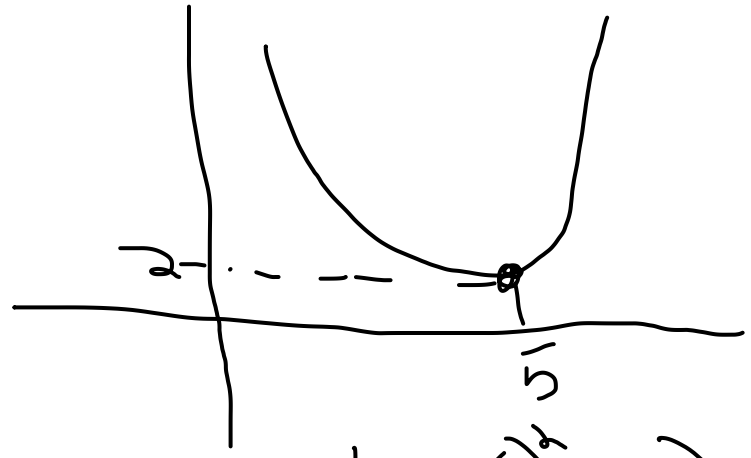
f(x) = 2x - 7
f(a) = 5

what is the value of a?

$f(a) = 2a - 7$
$f(a) = 5$
$2a - 7 = 5$
$2a = 5 + 7$
$2a = 12$
$a = \frac{12}{2}$
$\boxed{a = 6}$

Section3 Completing the Square.

* converting quadratic form into vertex form
* identify the vertex of a quadractic
* identify if its a max or a min
* analyze graphs



$y = (x-h)^2 + 2$

-ve means shift right
+ve means shift left

Standard form:
f(x) = ax^2 +bx + c

Vertex form:
f(x) = a(x-h)^2 + k (where (h, k) is the vertex)
* The vertex can be a minimum or a maximum

Example: (completing the square)
y = 2x^2 +12x + 16
a = 2, b = 12, c = 16

divide by two, square it,
add and subtract within the parentheses

y = 2(x^2 + 6) + 16
y = 2(x^2 + 6 + 9 - 9) + 16
y = 2((x^2 + 6 + 9) - 9) + 16
y = 2((x+3)^2 - 9) + 16
y = 2(x+3)^2 - 18 + 16
y = 2(x+3)^2 - 2

+ve means up
-ve means down

(a+b)^2 = a^2 + 2ab + b^2

vertex: (-3, -2)
max or min: minimum

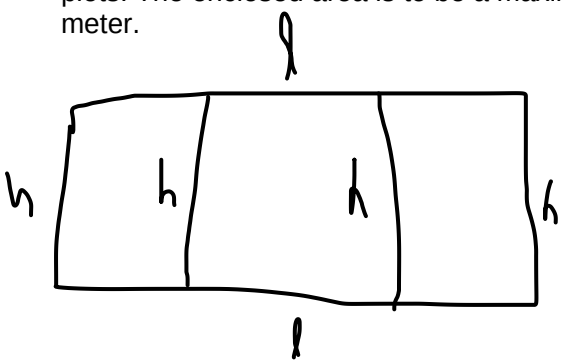
Done

Complete the square for the following:
a) y = x^2 + 4x + 1
b) y = -2x^2 + 12x + 7
c) y = 0.5x^2 - 3x + 6
d) y = 3x^2 + 15x - 5

State the following for the question above:
- the vertex
- max or min

Section 4: Applications of vertex form

A farmer has 5000 meters of fencing to enclose a rectangular field and subdivide it into three equal plots. The enclosed area is to be a maximum. Determine the dimensions of one plot of land, to the nearest meter.



2l +4h = 5000
A = l * h

$2l + 4h = 5000$ (1)
$A = l * h$ (2)

(1)
$L = (5000 - 4h) / 2$
$L = 2500 - 2h$

(1) -> (2)

$A = (2500 - 2h) * h$
$A = 2500h - 2h^2 + 0$
$A = -2h^2 + 2500h + 0$

$A = 2(h - 625)^2 + 781{,}250$

h=625 gives us the maximum area
$2L + 4h = 5000$
$2L + 4(625) = 5000$
$2L + 2500 = 5000$
$2L = 2500$
$L = 1250$ dimension: 1250 x 625

31st July

A club sells T-shirts for a fundraiser. If they charge $10, they sell 50 shirts. For every $0.50 increase in price, 2 fewer shirts are sold.

a) Write a revenue function.

b) What price should they charge to maximize revenue?

ANS) # t-shirts sold: $50 - 2x$
price per shirt: $10 + 0.5x$

revenue: $(50-2x)(10+0.5x)$

$r = -x^2 + 5x + 500$ $x = -2.5$

$r = -(x+2.5)^2 + 56.25$

how much are the tshirts selling for
how many tshirts are sold?
total revenue?

[redacted]

solving quadratic equations:

- solving quadratic eqns using factoring
- use the discriminant to determine the number of real roots
- solve systems of eqns involving linear and quadratic fxns

Solving by factoring
- setting the eqn to zero
- solve for x

x